

Muscular & Skeletal Systems

Bone is living tissue that answers to hormones. Muscle is a machine that runs on calcium.

BY THE END OF THIS PAGE YOU SHOULD BE ABLE TO

- Describe bone as living tissue, and name the two cells that rebuild it
- Explain how the skeleton protects the calcium level in blood
- Compare skeletal, cardiac, and smooth muscle
- Walk through how a muscle shortens, and the roles of calcium and ATP
- Explain how one muscle can make a weak twitch or a strong steady pull

1 Bone is not scaffolding

It is rebuilt all the time, and it is the body's calcium bank.

Two cells run a permanent renovation. **Osteoblasts** build bone. **Osteoclasts** break it down.

In a healthy adult the two roughly balance. You replace much of your skeleton over about ten years.

Tip the balance and the bone changes. Weight-bearing exercise favors the builders, so bone thickens. Bed rest favors the breakers, so bone thins. Aging and hormone loss do the same, and that is how osteoporosis develops.

THINK OF IT THIS WAY

Bone is less like a steel beam and more like a road under constant repair. Its strength depends on which crew is ahead.

2 The calcium loop, and why bone loses

Blood calcium is protected more fiercely than the skeleton is.

- 1 **Blood calcium drops.** The parathyroid glands notice.
- 2 **Parathyroid hormone is released.** This is the calcium-raising signal.
- 3 **Three things act.** Osteoclasts pull calcium out of bone. The kidney keeps calcium out of urine. The gut absorbs more, with help from vitamin D.
- 4 **Blood calcium returns to normal** — but bone was spent to do it.

That priority is not an accident. Calcium is needed for every nerve signal, every heartbeat, and every muscle contraction.

The body will drain the skeleton for years rather than let blood calcium drift, because a wrong calcium level stops the heart in minutes.

WHY THE NURSE CARES

This is why long-term vitamin D deficiency and chronic kidney disease damage bone. Neither starts as a bone disease. Both break the calcium loop, and the skeleton pays.

3 Three muscles, three different jobs

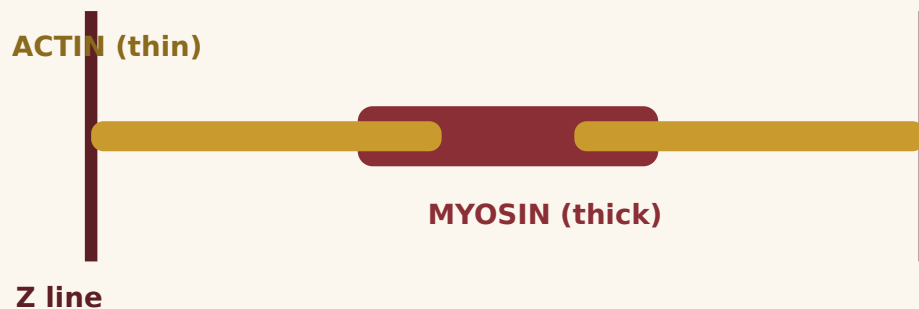
	SKELETAL	CARDIAC	SMOOTH
Control	You choose	Automatic	Automatic
Look	Striped, many nuclei	Striped, branched	Not striped, spindle shaped
Fires on its own?	No, it needs a nerve	Yes, it has its own pacemaker	Some of it does
Where	Attached to bones	Heart wall only	Vessels, gut, airways, bladder, uterus

Cardiac muscle is the one to notice. It makes its own rhythm, and its cells are joined electrically, so the whole chamber squeezes as one unit whether or not a nerve says so. That is why a transplanted heart still beats.

4 Sliding filaments — where force comes from

Drag the slider and watch force change with overlap alone.

LENGTH-TENSION: OVERLAP IS EVERYTHING



Maximum force. Every myosin head has thin filament within reach, and nothing is in the way. This is the length a muscle naturally rests at — the body keeps it near its best working point.

Muscle does not shorten by squeezing. Thick filaments grab thin filaments, pull them past, let go, and grab again.

Nothing changes length. The filaments only slide, and the ends of the unit are pulled closer together.

- 1 **A nerve signal arrives** and releases acetylcholine onto the muscle.
- 2 **The muscle fiber depolarizes**, and the signal spreads deep inside it.
- 3 **Calcium is released** from storage inside the cell. This is the trigger.
- 4 **Calcium uncovers the binding sites** on the thin filament, so the heads can attach.
- 5 **ATP powers the stroke.** A second ATP is needed to *let go* of each head.

- 6 **Calcium is pumped back** into storage, the sites are covered again, and the muscle relaxes.

Two ATP facts worth keeping. Relaxing costs energy too, not just contracting. Without ATP the heads stay stuck, and that is exactly what rigor mortis is. And pumping calcium back into storage is itself active transport, so a tired muscle relaxes slowly as well as pulling weakly.

5 From a twitch to a useful pull

One signal produces almost nothing. Real movement is built by repeating.

Stimulate a fiber once and you get a **twitch**.

Stimulate it again before it fully relaxes, and the second contraction builds on the first. That is **summation**.

Stimulate fast enough and the twitches blend into one smooth, steady pull. That is **tetanus**, and everything you do with your hands is tetanus.

Strength is also set by **how many motor units** are switched on. Lifting a pen calls on a few. Lifting a chair calls on many.

Between speed and recruitment, one nervous system can drive the same muscle across a huge range of force.

THINK OF IT THIS WAY

A twitch is one clap. Tetanus is applause, with claps so close together that you hear one steady sound.

6 Check yourself

Five questions. The wrong answers explain themselves, so they are worth reading.

1. Blood calcium falls. What happens next?

- a) Parathyroid hormone rises, and osteoclasts release calcium from bone
- b) Parathyroid hormone falls, and bone takes in more calcium
- c) Osteoblasts release calcium into blood
- d) The kidney puts more calcium into urine

2. Which feature belongs only to cardiac muscle?

- a) It is striped
- b) You control it voluntarily
- c) It makes its own rhythm and its cells contract as one unit
- d) It contains actin and myosin

3. What is the direct trigger that lets the filaments grab each other?

- a) Acetylcholine landing on the muscle membrane
- b) A rise in calcium inside the muscle fiber
- c) ATP binding the myosin head
- d) Sodium entering the fiber

4. Why does rigor mortis happen?

- a) Calcium is pumped away permanently
- b) There is no ATP left to release the heads from the thin filament
- c) The muscle runs out of acetylcholine
- d) The filaments shorten and cannot lengthen again

5. How does the nervous system make one muscle produce a smooth, strong pull instead of a twitch?

- a)** It sends a bigger action potential
- b)** It stimulates fast enough that twitches blend, and switches on more motor units
- c)** It moves calcium from the blood into the muscle
- d)** It stretches the muscle first

Answer key

1. **Parathyroid hormone rises, and osteoclasts release calcium from bone**

Correct. Bone, kidney, and gut all act to raise blood calcium, and bone pays for it.

2. **It makes its own rhythm and its cells contract as one unit**

Right. Its own pacemaker and joined cells are why a transplanted heart beats.

3. **A rise in calcium inside the muscle fiber**

Correct. Calcium uncovers the binding sites, and nothing can attach until it does.

4. **There is no ATP left to release the heads from the thin filament**

Right. Letting go costs ATP, so with no ATP the grips stay locked.

5. **It stimulates fast enough that twitches blend, and switches on more motor units**

Correct. Speed plus recruitment covers a huge range of force.

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Joyce University of Nursing and Health Sciences

Course resources by **Dr. Vidya Garg**